

APPENDIX A

A.1 Explanation of the analytical approach: step-by-step example

Before delving into the analytical approach, it is essential to understand how it functions and to grasp the dynamic total time formulation outlined in Section 2.2.3.1 in (eq. A1). This equation deterministically encapsulates the logic for tracking station states, robot manipulator movements, and parallelizable overlaps within the Unidirectional Streamflow Control (U-SFC) framework.

$$\Delta t_{\text{effective}}(t) = \sum_{k \in K} \sum_{i \in S} \left[T_{k,i} \cdot v_{k,i}(t) \cdot 1_{\{\forall S_j \in S \setminus \{S_i\}, \text{State}(S_j,t)=0\}} \right. \\ \left. + \max \left(0, T_{k,i} \cdot v_{k,i}(t) - \max_{S_j \in S \setminus \{S_i\}} \{T_{k,j}^{\text{remaining}}(t)\} \right) \right. \\ \left. \cdot 1_{\{\exists S_j \in S \setminus \{S_i\}, \text{State}(S_j,t)=1\}} \right] + M_{\text{travel}}(t) \cdot 1_{\{\forall S_l \in S, \text{State}(S_l,t)=0\}} \quad (\text{eq. A1})$$

By applying these principles to a simplified scenario involving two trays, it becomes possible to visualize how discrete events are processed over time, how station availability is tracked, and how overlapping tasks are handled within the analytical framework.

To enable precise tracking and organization within the factory sequencing, a unique identifier (ID) is assigned to each production tray. This nomenclature systematically combines the core logistics parameters of the manufacturing run, following the structure: *Invoice order_Article code_Batch size_Tray number out of the total*.

To ground this in the experimental data, Table A1 presents the processing sequences of two trays with different operation times: 6075170_6...100_230_1/2 and 6075170_6...100_230_2/2. Both trays follow the same route (B and DF) but differ significantly in processing times. The following example describes, step by step, how the analytical model processes these two trays and determines when time should be added to T_{total} .

Table A1. 6075170_6...100_230_1/2 and 6075170_6...100_230_2/2 trays manufacturing route

Tray ID	S_1	S_1 Time	S_2	S_2 Time
6075170_6...100_230_1/2	1	400	1	400
6075170_6...100_230_2/2	1	60	1	60

When the approach begins processing the first tray (6075170_6...100_230_1/2), the robot collects it from the input station and transports it to B, changing its state to "busy". However, before proceeding to add time to T_{total} , the approach checks the status of all other stations. Since none are currently marked as "busy", both the robot movement time (M_{travel}) and the 400 seconds of brushing ($T_{6075170_6...100_230_1/2,B}$) are added to T_{total} (see Figure A1).

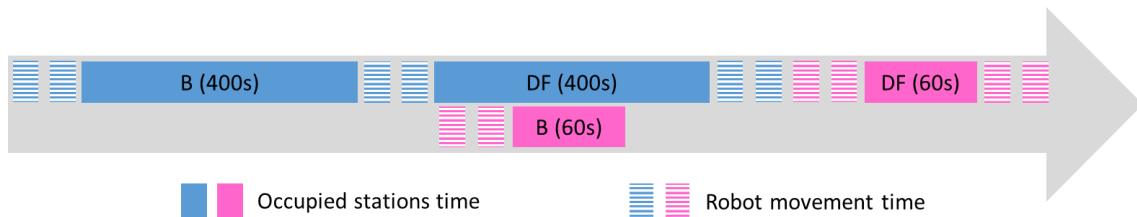


Figure A1. Timeline representation of station occupancy and robot movements during tray processing. Only non-overlapping durations are added to T_{total} . Overlapping processing times and transport actions are excluded.

Only non-overlapping durations are added to T_{total} , while overlapping processing times and transport actions are excluded. In this context, 'transport actions' refers to the complete logistical cycle of picking, transferring, and placing trays between the different stations of the FMC. As visualized in Figure A1, robot movement times (M_{travel}) often appear in consecutive blocks. This occurs because the robot executes movements in pairs: the first block represents the travel to a station to pick up a tray, and the second represents the transport and placement at the destination. Each of these specific segments (picking and placing) has its own duration calculated based on the actual distances within the experimental cell.

Once brushing is completed, the station switches to "finished", and the robot picks up the tray and transfers it to DF. As DF is in an "idle" state, it transitions to "busy", and again both the robot movement $M_{B,DF}$ and the 400 seconds of processing ($T_{6075170_6\dots100_230_1/2,DF}$) are added to T_{total} , since no other station is currently in a "busy" state.

While DF is still processing the first tray, the approach introduces the second tray (6075170_6...100_230_2/2) into B. As always, the approach first checks all other stations' states. DF remains "busy", so robot movements times are not added to T_{total} . Then, the approach compares the 400 seconds remaining at DF with the 60 seconds required for brushing ($T_{6075170_6\dots100_230_2/2,B}$). Since the entire brushing operation is completed in parallel with the ongoing DF task, B changes to "busy" but no additional time is added to the global total. In the hypothetical case where the brushing time exceeded the remaining time at DF, the difference would be added to T_{total} , consistent with the term $(\max(0, T_{k,i} \cdot v_{k,i}(t) - \max_{S_j \in S \setminus \{S_i\}} \{T_{k,j}^{\text{remaining}}(t)\}))$ in the dynamic total time formulation.

Once brushing is finished, B switches to "finished"; however, since DF is still "busy", the robot must wait. After the 400 second process at DF is complete, the station status becomes "finished", the robot transfers the first tray to VI, and DF changes to "idle". As no other stations are "busy" at that point, both robot movements added to T_{total} .

From this point onward, with only the second tray being processed and no overlapping operations, all remaining processing and movement times are fully added to T_{total} , as nothing else can occur in parallel. Although this example uses only two trays for clarity, the approach is designed to handle larger batches with up to four stations that possibly could work simultaneously. Management of station states ("idle", "busy", and "finished") is essential to ensure that the global time only increases when operations cause actual delays.

A.2 Application of the analytical approach

With the foundational understanding of the analytical approach established through the step-by-step example, the practical application of the model follows. To facilitate verification and replication of the model's functionality, a simulated dataset has been prepared, closely mirroring the experimental production assay while incorporating necessary modifications for confidentiality. For confidentiality purposes, all production times in the dataset have been standardized to 2 seconds per tool, regardless of the actual variability observed in real operations. Additionally, sensitive identifiers (such as full article codes, exact batch sizes, and invoice orders) have been partially anonymized or aggregated. The input data is derived from *Experimental_production_orders.xlsx*, a real-world production log encompassing 11 trays from January 2024.

The accompanying Python script (*Analytical_model_by_invoice_order.py*) implements the DES. Readers are encouraged to execute the script in a Python environment with pandas and numpy installed, using the provided Excel file (placed at the hardcoded path).

A.3 Simulation results and insights

The analytical approach delivers valuable metrics for FMC optimization, including working times and utilization percentages for each station and the robot, alongside the overall cycle time. As can be seen in Table A2, execution of the provided Python script on the example dataset (Experimental_production_orders.xlsx) generates the following synthetic results (derived from anonymized and standardized inputs for confidentiality), rather than real production data. The total cycle time (T_{total} ,) stands at 1.25 h.

Table A2. Output from dataset simulation

Resource	Station total working time	Utilization rate
Brushing (102)	0.71 h	56.36%
Drag finishing (103)	0.26 h	20.87%
Painting (104)	0 h	0%
Laser marking (105)	0.44 h	35.45%
TX200 robot	64.64 h	64.44%

These metrics pinpoint key inefficiencies (such as the robot's heavy workload) while the script's flexible architecture (e.g., adjustable robot movement times or task queuing rules) supports effortless customization for varied scenarios.